Week 8: Financing Multinationals
Currency Swaps, Basis, and Liquidity

FINA3020 at CUHK Business School

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- **Part 1 (already in Week 5 slides):** taxes, remittance limits, transfer prices, a full project valuation, and governance.
- **Part 2:** choose the currency, market, and maturity for financing.
- Calculate borrowing cost after the hedge, fees, and executable dealer quotes.
- Case studies of actual management responses.
- How bank funding constraints and swap lines affect corporate access to cash.

International Capital Budgeting: Core Framework

[Core Concept]

Evaluating foreign capital expenditure (CAPEX) requires converting cash flows and discount rates across currencies consistently. Two valuation methods:

- **Home Currency Approach (Parent Perspective)**: Forecast foreign cash flows CF_t^*, translate each date-t flow using a consistently risk-adjusted exchange-rate forecast $E_0[S_t]$, and discount at a home-currency required return k_h appropriate for those cash flows.
- **Foreign Currency Approach (Subsidiary Perspective)**: Discount foreign cash flows CF_t^* at Foreign WACC k_f, convert resulting foreign NPV to home currency at current spot rate S_0.

The two methods agree only when project risk, FX risk, cash-flow forecasts, and currency-specific discount rates are priced consistently. UIP and relative PPP describe exchange-rate relations. Project WACCs also reflect operating and financial risk.

View Proof of Valuation Equivalence →

Example →

Parent vs. Subsidiary Cash Flows: Why They Differ [Core Concept]

Value cash flows and assets available to the parent after tax, remittance, financing, and capital-control constraints.

Why do subsidiary & parent cash flows differ?:

1. **Taxation Frictions**: Foreign corporate income tax vs. home tax credit ceilings.
2. **Remittance Restrictions**: Host government limits on dividend repatriations (blocked funds).
3. **Intra-Company Overhead**: Licensing fees, management charges, and transfer prices paid to parent.
4. **Exchange Rate Risk**: Spot rate movements between cash generation and dividend remittance.

View Step-by-Step 3-Year Project Example →

Example →

International Taxation & Foreign Tax Credits [Core Concept]

Host governments generally tax local subsidiary income. Home-country treatment varies: dividends may be exempt, credited, deferred, or subject to minimum-tax and anti-deferral rules.

- **Foreign Tax Credit (FTC)**: In a credit system, qualifying foreign taxes may offset home tax on foreign-source income, subject to jurisdictional or income-basket limitations and domestic-law eligibility rules.

$$\text{Usable FTC} \leq \text{Home tax attributable to the relevant foreign-source income}$$

- Two tax outcomes:
  - In a simplified worldwide credit system, a lower qualifying foreign tax can leave residual home tax.
  - If qualifying foreign tax exceeds the limitation, current home tax may be zero. Use of excess credits depends on domestic carryover rules.

View Detailed Tax Credit & Remittance Withholding Calculation →

Blocked Funds & Transfer Pricing Strategy

[Core Concept]

Blocked Funds (Remittance Limits)

- Host country capital controls restrict dividend transfers abroad.
- Unrepatriated cash may be reinvested locally, held in liquid assets, or released later. Focus on after-tax return.
- A restriction reduces value when the available local uses earn less than the parent could earn elsewhere, after adjusting for risk.

Tax credits and blocked funds → Example →

Transfer Pricing Strategy

- Setting internal prices for components, IP, or management fees.
- Transfer prices allocate income and customs value across entities, affecting tax and remittance outcomes in more than one jurisdiction.
- Prices must follow applicable arm's-length, documentation, substance, customs, and anti-avoidance rules.

Transfer Prices and Managerial Incentives [Core Concept]

- A transfer price assigns revenue and cost to units within a group. It affects reported divisional profit, tax, performance targets, and internal investment decisions.
- The arm's length principle asks what independent parties would charge in comparable circumstances. Applicable tax rules constrain the internal prices firms can use.
- Comparable transactions are harder to identify for unique intellectual property. Documentation, audits, and advance pricing agreements can reduce disputes.
- A manager rewarded on divisional profit may resist a group-optimal transfer price. Consolidated value requires including cash flows and taxes in every affected entity.

Source: OECD Transfer Pricing Guidelines (2022).

Example →

Transfer Pricing Direction

[Core Concept]

Hypothetical tax rates: German subsidiary 30%, US parent 20%. The subsidiary makes an item for EUR 80 and sells it to the parent, which resells for EUR 200.

Transfer price	German profit	US profit	Combined tax
EUR 100	20	100	$0.30(20) + 0.20(100) = 26$
EUR 140	60	60	$0.30(60) + 0.20(60) = 30$

- Raising the price moves EUR 40 of profit toward higher-tax Germany and increases tax by EUR 4. If instead the subsidiary buys from the parent, raising the transfer price moves profit in the opposite direction.

- Throughout this section on transfer pricing, I give hypothetical examples that may violate the law in some jurisdictions (some are strict on transfer pricing). For simplicity, I ignore withholding, residual tax, and customs.

Historical tax example → Historical royalties → Example →

From Operating Profit to Cash Flow

[Core Concept]

$$\mathrm{PV}_t = \frac{\Big(\overbrace{\mathrm{Rev} - \mathrm{Costs} - \mathrm{Dep}}^{\text{EBIT}} - \overbrace{\tau \times \mathrm{EBIT}}^{\text{Corp. tax}} + \mathrm{Dep} - \mathrm{CAPEX}_t - \Delta\mathrm{NWC}_t\Big) \times (1 - \mathrm{WHT}) \times \mathrm{FX}_t}{(1 + k_h)^t}.$$

$$\mathrm{FX}_t = \begin{cases} E_0[S_t] & \text{unhedged: convert at the expected spot rate} \\ F_{0,t} & \text{hedged: sell the euro amount forward at the forward rate} \end{cases} \quad \text{(USD per 1 EUR)}$$

- Use incremental cash flows. Depreciation is noncash: deduct it to compute tax, then add it back. The parentheses give free cash flow to the subsidiary.
- Withholding tax applies to the cash remitted, and the FX rate converts it to USD. A forward fixes $F_{0,t}$ but not the euro amount, so hedge the amount that you will have.

Example →

Example 1: Build the Subsidiary's Operating Cash Flow [Core Concept]

Year 1 operating inputs (EUR M): Revenue = 12.0, cash operating costs = 6.0, depreciation = 2.0, local corporate tax rate = 20%, and no incremental CAPEX or working-capital investment in Year 1.

Cash Flow Step	EUR Millions
Revenue	12.00
Less: Cash operating costs	(6.00)
Less: Depreciation	(2.00)
Earnings before tax	4.00
Less: Local corporate tax (4.00× 20%)	(0.80)
Net operating profit after tax	3.20
Add back: Depreciation (non-cash)	2.00
After-tax operating cash flow	**5.20**

Example 2: Convert and Discount Parent Cash Flows

[Core Concept]

Project assumptions: Initial CAPEX = EUR 10.0M; $S_0 = 1.10$ USD per 1 EUR; subsidiary free cash flows are EUR 5.2M, 6.4M, and 12.6M; all cash is remittable with no additional parent tax; expected future spot rates are 1.12, 1.14, and 1.16 USD per 1 EUR; home required return = 10%.

Year	EUR Cash Flow	$E_0[S_t]$	USD Parent Cash Flow	Present Value at 10%
0	(10.000)	1.10	(11.000)	(11.000)
1	5.200	1.12	5.824	5.295
2	6.400	1.14	7.296	6.030
3	12.600	1.16	14.616	10.981

$$\mathrm{PV}_t = \frac{CF_t^{\mathrm{EUR}} \times E_0[S_t]}{(1+k_h)^t}, \qquad \mathrm{NPV}_{\mathrm{USD}} = -11.000 + 5.295 + 6.030 + 10.981 = +\mathbf{\$11.306}\mathrm{M}$$

Baseline Decision

The project creates value before new remittance restrictions: **accept**.

Example 2b: Hedge the Parent Cash Flows with Forwards [Core Concept]

Same project as Example 2. Hold EUR cash flows at the stated amounts and use a common 10% hurdle rate for this scenario comparison. Sell those amounts forward at $F_{0,1} = 1.11$, $F_{0,2} = 1.12$, $F_{0,3} = 1.13$ USD per EUR.

Year	EUR Cash Flow	Forward Rate	Hedged USD Cash Flow	Present Value at 10%
0	(10.000)	1.10	(11.000)	(11.000)
1	5.200	1.11	5.772	5.247
2	6.400	1.12	7.168	5.924
3	12.600	1.13	14.238	10.697

$$\mathrm{PV}_t = \frac{CF_t^{\mathrm{EUR}} \times F_{0,t}}{(1+k_h)^t}, \qquad \mathrm{NPV}_{\mathrm{hedged}} = -11.000 + 5.247 + 5.924 + 10.697 = +\$\mathbf{10.868}\mathrm{M}.$$

- The hedged scenario's PV is lower by approximately \$0.437M, using unrounded values. Economic valuation also requires discount rates consistent with each cash flow's risk.
- For the stated EUR cash flows, hedged receipts are fixed. At future spot rates of 1.00 throughout, unhedged NPV is +\$8.483M; at 1.24 throughout, it is +\$13.159M.

Example 3: Adjust Only Cash Reaching the Parent [Core Concept]

Shock to Year 2 cash flow: EUR 6.4M faces a 20% dividend-remittance withholding tax and a one-year freeze. Blocked funds earn 3% locally and are released in Year 3; $E[S_3] = 1.16$ USD per 1 EUR and home WACC = 10%.

Baseline: immediate Year 2 remittance

$$PV_0 = \frac{6.4(1.14)}{1.10^2} = \$\mathbf{6.030}\text{M}$$

Withholding only (before any FTC)

$$PV_0 = \frac{6.4(0.80)(1.14)}{1.10^2} = \$\mathbf{4.824}\text{M}$$

Withholding plus one-year freeze

$$PV_0 = \frac{6.4(0.80)(1.03)(1.16)}{1.10^3} = \$\mathbf{4.596}\text{M}$$

Value lost on the Year 2 cash flow

$$6.030 - 4.596 = \$\mathbf{1.434}\text{M}$$

Comprehension Check

When withholding tax qualifies for a credit, total tax equals withholding tax plus home-country tax after usable foreign tax credits.

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Three-Year Project: A Full Repatriation Calculation [Core Concept]

All operating amounts are EUR millions. Initial investment is EUR 30 at EUR/USD $= 1.10$. Depreciable basis is EUR 27, annual depreciation EUR 5, and terminal book value and salvage both EUR 12. The remaining initial EUR 3 is a nonrecoverable setup cost.

Year	Revenue	Cash cost	Taxable profit	After-tax cash
1	30	18	7	10.25
2	32	19	8	11.00
3	34	20	9	11.75 + 12 salvage

- Assume host corporate tax 25%, 10% withholding on all modeled distributions including terminal cash, no additional home tax, and full remittance.
- In reality, capital returns and dividends may receive different treatment.

Year 1 calculation → Year 2 calculation → Year 3 calculation →

Three-Year Project: Parent NPV and FX Sensitivity [Core Concept]

Year	After withholding, EUR M	Converted USD M	PV at 10%, USD M
1	9.225	10.1475	9.2250
2	9.900	10.8900	9.0000
3	21.375	23.5125	17.6653

$$\text{NPV} = 9.225 + 9 + 17.6653 - 30(1.10) = \text{USD } 2.8903\text{M}.$$

- If initial FX stays 1.10, but every future conversion is 1.05, NPV becomes $35.8903(1.05/1.10) - 33 = \text{USD } 1.2589\text{M}$.
- If every future conversion is 1.15, NPV becomes USD 4.5217M.
- The sequence is operating cash flow, local tax, remittance restrictions, withholding and usable tax credits, currency conversion, then discounting.

Transfer Pricing and Consolidated Project Value

[Core Concept]

Continue the three-year exercise. In Year 1 only, the subsidiary pays an additional EUR 4M to the parent for inputs. Assume no incremental real cost, a 20% parent tax on that markup, and no tax credit or other tax interaction.

Incremental effect	EUR M
Subsidiary tax saved: 25% of 4	+1.00
Subsidiary cash before withholding	-3.00
Reduced remittance after 10% withholding	-2.70
Parent markup after 20% tax	+3.20
Consolidated incremental parent cash	+0.50

$$\Delta \mathrm{NPV} = 0.50(1.10)/1.10 = \mathrm{USD}\ 0.50\mathrm{M}.$$

- The subsidiary's remittance PV falls from 9.225 to 6.525, but that omits the parent's input-sale income. The consolidated calculation includes both.

Full transfer-price comparison →

Corporate Governance across Borders

[Core Concept]

- Shareholders delegate decisions to managers. Distance, language, local regulation, and complex group structures make monitoring costly.
- Parent and affiliate managers can disagree over investment, transfer pricing, and retention of cash. Controlling shareholders can also conflict with minority owners.
- Boards oversee strategy and management. Independent directors, local expertise, internal controls, external audits, and enforceable disclosure work together.
- Cross-listing can broaden the investor base and impose additional disclosure. Governance benefits depend on the listing program, enforcement, and actual oversight.

Financial statements → MNC examples →

Company comparisons →

Example →

Tunneling, Disclosure, and Enforcement [Core Concept]

- Tunneling transfers value from a firm to its controller through unfair related-party sales, loans, guarantees, or dilution. Ordinary arm's length intra-group trade is different.
- The Samsung case in Bae, Kang, and Kim (2002) illustrates the risk of transfers that may favor a controlling family.
- Enron illustrates concealed liabilities and misleading earnings through complex entities.
- The Renault–Nissan alliance illustrates conflicts among partners and public shareholders. Volkswagen and Siemens illustrate the importance of oversight and cross-border enforcement.
- Legal-origin research reports historical associations between institutions and finance. Assess current shareholder rights, enforcement, and firm governance when evaluating financing costs.

Example →

Agency Problems inside a Multinational Group [Core Concept]

- **Shareholders and managers**: managers may expand the group, retain cash, or pursue private benefits even when shareholders would prefer a payout.
- **Headquarters and affiliate managers**: local managers know more about customers and costs. Headquarters must distinguish weak effort from adverse local conditions.
- **Controlling and minority shareholders**: a controller can benefit another affiliated company through an unfair price, loan, guarantee, or share issue.
- **Creditors and shareholders**: risky investment or asset transfers can reduce the value of existing debt. Guarantees can move exposure across entities.

Renault–Nissan → Enron → Samsung and tunneling →

Monitoring and Boards across Jurisdictions

[Core Concept]

- Boards oversee strategy, senior management, and major related-party transactions. Managers run daily operations.
- Distance, language, entity boundaries, and differences in disclosure complicate monitoring. Local knowledge and independent judgment are both valuable.
- Useful controls include consolidated reporting, independent audits, review of guarantees and related-party transactions, and clear escalation of compliance concerns.
- Board effectiveness depends on incentives, information, expertise, and enforcement.

Vale and monitoring → Volkswagen oversight →

Investor Protection, Cross-Listing, and Enforcement [Core Concept]

- Investor protection includes voting and disclosure rights, remedies for abusive transactions, and credible enforcement. Written rights and practical access to remedies can differ.
- Legal-origin research compares historical country patterns, such as those in the UK and France. Evaluate each firm's shareholder rights, enforcement, and oversight under current rules.
- Cross-listing places shares or depositary receipts in another market. An ADR represents an interest in foreign shares, and program types carry different reporting obligations.
- A broader investor base and stronger disclosure can lower financing frictions. Listing costs and compliance obligations also rise, and misconduct remains possible.

Petrobras ADRs → Legal origins → Siemens enforcement →

Multinational Investment: The Complete Decision [Core Concept]

- **Entry**: compare exporting, independent contracting, and owned production using trade costs, scale, local demand, and control requirements.
- **Funding**: compare the amount, currency, and location of borrowing, internal funds, and equity. Financing frictions can change feasible investment.
- **Value**: calculate incremental operating cash, taxes, remittances, terminal cash, FX conversion, and discounting from the parent's perspective.
- **Governance**: establish who makes decisions, who bears risk, and who can transfer value across entities. Group accounts combine results across entities, while managers and owners can still have conflicting incentives.

Example →

Week 5 Decisions Become Treasury Tasks

[Core Concept]

Week 5 concept	Week 8 application
Foreign project NPV	Can treasury fund the investment in the currency and on the dates it needs?
Cost of capital and debt location	Compare local debt with parent borrowing plus a currency swap.
Parent versus subsidiary cash	Identify cash that can actually move between entities.
Working capital	Fund inventory and receivables until customers pay.
Distress and financing frictions	Test a failed rollover or a collateral call before committing capital.

Funding a Euro-Earning Subsidiary

[Core Concept]

- **Local EUR loan:** service debt from EUR receipts. Check local rates, covenants, collateral, and refinancing dates.
- **Parent USD borrowing plus a currency swap:** use the parent's funding access while creating EUR debt service.
- **Parent equity or an intercompany loan:** use internal cash, subject to transfer, tax, and repayment constraints.
- Compare the **same net proceeds, currency, maturity, and risk**. A lower foreign coupon alone is insufficient.

Corporate issuance evidence →

Three Contracts with Different Jobs

[Core Concept]

Instrument	Cash flows	Typical treasury use
FX swap	Exchange principal on two agreed dates.	Bridge a temporary currency funding gap.
Cross-currency interest-rate swap	Exchange currency cash flows, usually principal and periodic interest.	Transform the debt service on a multi-year loan or bond.
Central-bank swap line	Two central banks exchange currencies. The receiving central bank provides local dollar liquidity.	Indirect support through eligible banks and funding markets.

A corporate treasury negotiates its own bank facilities and derivatives.

An FX Swap Matches Two Currency Dates

[Core Concept]

A firm owes EUR 1M on day 10 and receives EUR 1M on day 20. A forward-starting FX swap can buy EUR on day 10 and sell EUR on day 20.

Date	EUR cash flow from swap	USD cash flow from swap
Day 10	+1,000,000	$-1{,}000{,}000F_{0,10}$
Day 20	-1,000,000	$+1{,}000{,}000F_{0,20}$

- The incoming and outgoing EUR amounts now match the commercial cash flows. A spot-to-day-30 swap would leave timing gaps.
- A standard FX swap exchanges principals on near and far dates. A cross-currency interest-rate swap can also exchange periodic interest and longer-term principal.
- Fixing both exchange rates removes that exchange-rate uncertainty for matched flows. Rollover, liquidity, counterparty, and settlement risks remain.

Two-date cash-flow table →

The Benchmark Covered Interest Parity

[Core Concept]

Quotation convention: BASE/QUOTE expresses the price of 1 unit of BASE currency in QUOTE units.

- Let S be the spot rate and F the forward rate for the pair foreign/home: home currency H per unit of foreign currency (BASE = foreign, QUOTE = home)
- Let i_H be the home (QUOTE) nominal interest rate and i_F the foreign (BASE) nominal interest rate, both annual, over year fraction t
- Theoretical CIP forward rate F^*:

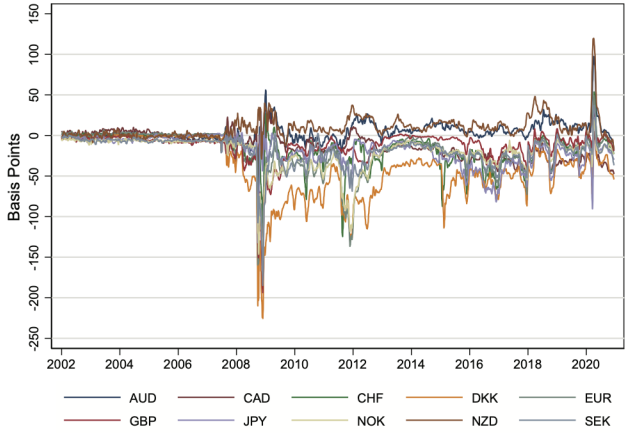
$$F^* = S \times \frac{1 + i_H t}{1 + i_F t}$$

- Benchmark assumption: matched counterparties can borrow/lend at the quoted rates and trade spot/forward simultaneously without balance-sheet, credit, collateral, or execution frictions

Mathematical Derivation →

Post-2008 CIP Breakdown

[Core Concept]



Source: Cerutti, Obstfeld, Zhou (2021)

Empirical Data & Cerutti et al. (2021) →

- Pre-2008: deviations for major-currency benchmark instruments were usually small and often within estimated transaction-cost bounds
- Post-2008: persistent negative quoted bases emerged for EUR, JPY, and GBP against USD
- Relative to the chosen benchmark rates, synthetic dollar funding via FX swaps has often cost more than direct dollar funding

Defining the Cross-Currency Basis (Δ and b)

[Core Concept]

Here S and F are **USD per unit of foreign currency**. t is the year fraction. Use comparable maturities and rate conventions.

$$F^* = S\frac{1 + i_{USD}t}{1 + i_F t}.$$

1. Borrow foreign currency, convert it to dollars spot, and cover the full foreign repayment forward:

$$i_{USD}^{syn} = \frac{(1 + i_F t)F/S - 1}{t}.$$

2. Define the annualized funding wedge $\Delta = i_{USD}^{syn} - i_{USD}$. A positive Δ means synthetic dollars cost more. This occurs exactly when $F > F^*$.

3. If quoted basis b is added to the foreign interest leg, then

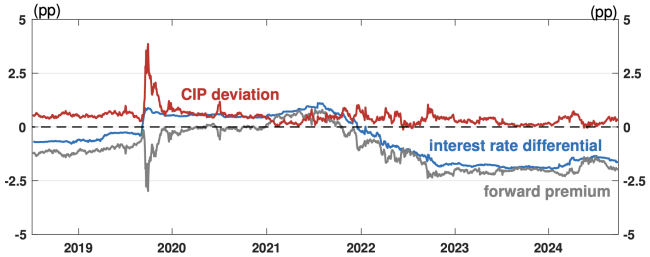
$$1 + i_{USD}t = (1 + (i_F + b)t)F/S, \qquad b = -\Delta S/F.$$

Thus dollar scarcity can appear as $\Delta > 0$ or $b < 0$: the sign depends on the definition.

Quote units and basis signs →

Korea: The Cost of a 47-Basis-Point Wedge

[Core Concept]



Source: Kim and Kim (2025). Bank of Korea. Historical three-month USD/KRW exhibit. Cost inputs below are illustrative.

The wedge is persistent

- Positive on most days in the displayed historical series
- Cost exercise: **47 bps**, or **386 bps** under stress

Extra cost on a $100m, 3-month hedge

$$\$100m \times \Delta \times \frac{3}{12}$$

$117,500 at 47 bps
$965,000 at 386 bps

Investors with cheap USD funding may earn the wedge. Insurers and asset managers can pay it as a hedging or funding premium.

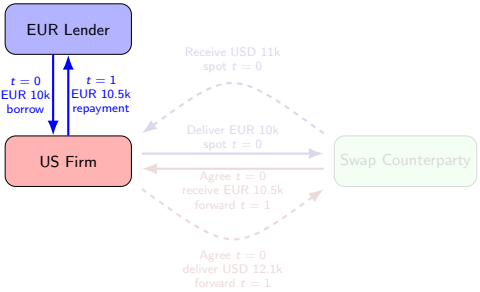
If the arbitrage is visible almost every day, why does it survive?

Basis points to dollars →

Synthetic Dollar Borrowing: The Complete Repayment

[Core Concept]

Spot EUR/USD $= 1.10$. One-year EUR and USD rates 5% and 10%, zero spreads. The firm borrows EUR but wants USD.



- Benchmark forward: $F = 1.10(1.10/1.05) = 1.15238095$.
- Sell EUR 10,000 spot and receive USD 11,000.
- Hedge the full EUR repayment: $10{,}000(1.05) = 10{,}500$.
- Pay $10{,}500F =$ USD 12,100 to receive the EUR needed by the lender.

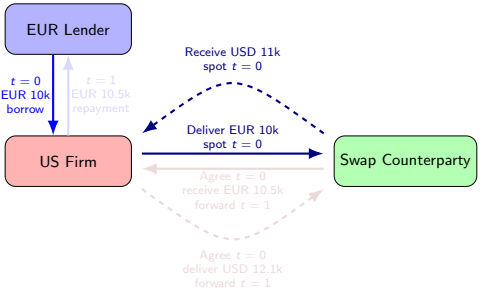
The covered USD cost is $12{,}100/11{,}000 - 1 = 10\%$.

Principal and interest → Borrower spreads →

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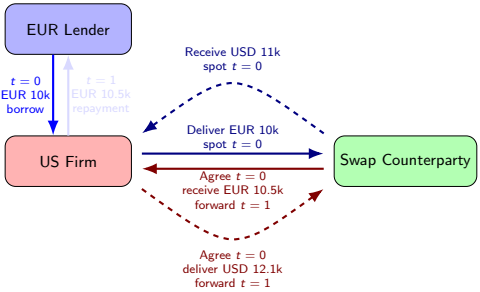
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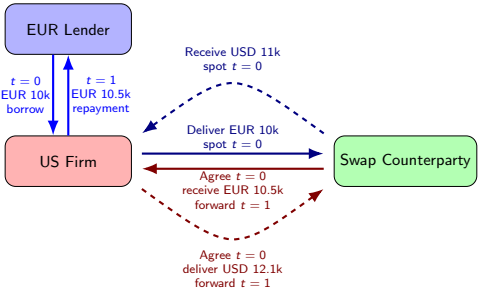
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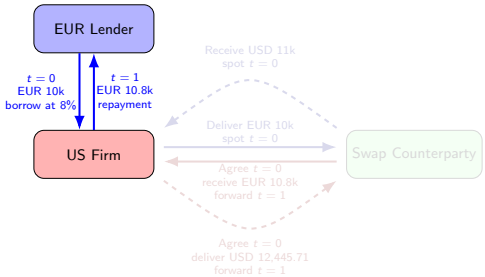
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Principal and interest → Borrower spreads →

Borrower Spreads Can Change the Cheapest Currency [Core Concept]

Benchmark rates: EUR 5%, USD 10%, so $F = 1.15238095$. The firm borrows EUR at 8% or USD at 16%.



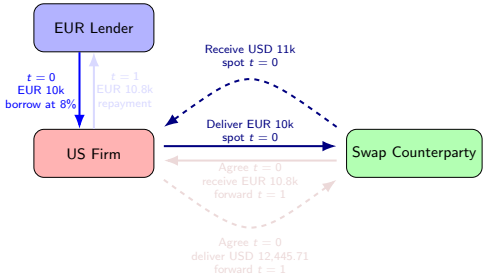
- Borrow EUR 10,000 at the firm's 8% rate. Receive USD 11,000 spot.
- EUR repayment: $10{,}000(1.08) = 10{,}800$.
- Forward-covered USD repayment: $10{,}800F = 12{,}445.71$.
- Direct USD repayment: $11{,}000(1.16) = 12{,}760$. Saving: **USD 314.29**.

The covered USD cost is 13.1429%. The saving reflects the firm's different borrowing spreads.

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[Core Concept]

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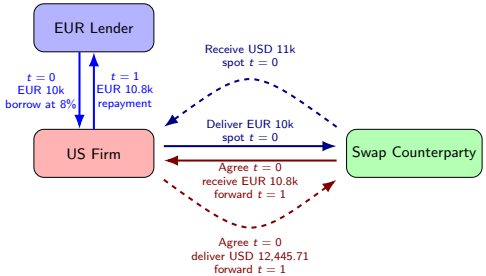
- Borrow EUR 10,000 at the firm's 8% rate. Receive USD 11,000 spot.
- EUR repayment: $10{,}000(1.08) = 10{,}800$.
- Forward-covered USD repayment: $10{,}800F = 12{,}445.71$.
- Direct USD repayment: $11{,}000(1.16) = 12{,}760$. Saving: **USD 314.29**.

The covered USD cost is 13.1429%. The saving reflects the firm's different borrowing spreads.

Borrower Spreads Can Change the Cheapest Currency

[Core Concept]

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A Dollar Bond Can Finance Euro Debt Service

[Core Concept]

Illustration. A US parent needs EUR 10M for five years. Assume spot is 1.10 USD per EUR.

Date	USD bond	Currency swap
Today	Receive USD 11M.	Pay USD 11M and receive EUR 10M.
Each year	Pay USD 0.66M interest (6%).	Receive USD 0.66M and pay EUR 0.45M (4.5%).
Year 5 principal	Repay USD 11M.	Receive USD 11M and repay EUR 10M.

Combined financing: receive EUR 10M, pay EUR 0.45M annually, and repay EUR 10M. Compare with a direct EUR loan at the same maturity.

The Swap Must Match the Financing [Core Concept]

- **Match every payment:** notional, amortization, coupon dates, day counts, and final maturity.
- **Fixed or floating:** a floating EUR payment leaves EUR interest-rate risk. A fixed EUR payment locks that rate for the agreed term.
- **Early exit:** cancelling a project can leave both the bond and a costly swap closeout.
- **Counterparty and collateral:** check replacement cost, collateral terms, and the cash needed during adverse market moves.
- **Decision:** compare total EUR cash payments and liquidity commitments with the direct EUR offer.

Newell: Robert Westreich's Financing Response [Core Concept]

Historical case, reported in May/June 2023. Westreich was Senior VP, Treasurer and Chief Tax Officer of Newell Brands.

- **Situation:** refinancing became more expensive. Seasonal inventory purchases also required short-term borrowing before customer receipts.
- **Action:** Newell used cross-currency swaps to convert dollar borrowing into euro obligations.
- **Reported result:** Westreich described a 200 bp benefit on bonds with an average 6.5% coupon.
- **Analysis:** identify the EUR exposure and final repayment before interpreting that interest saving as an increase in firm value.

Questions for the treasurer →

Newell: The Exposure behind the Swap [Core Concept]

- The 2023 annual filing identifies two USD 500M cross-currency swaps entered in 2022, maturing in September 2027 and 2029.
- Newell designated them as **net investment hedges** of subsidiaries with EUR functional currencies and EUR net assets.
- On those swaps, Newell paid floating EUR interest and received floating USD interest. EUR interest-rate exposure therefore still matters.
- The filing reported no derivatives requiring collateral before settlement. The margin-call exercise later in this lecture assumes a different contract.

Week 5 connection: financing decisions depend on the foreign business being financed and the exposure management wants to retain.

All-in Cost of Synthetic Dollar Borrowing

[Core Concept]

Illustrative offers. Raise USD 100M for one year. Direct USD loan: 5.60%. EUR loan: 3.20%.

$$S = 1.1000, \quad F = 1.1220 \quad \text{(USD per EUR)}.$$

1. Borrow $100/1.10 =$ EUR 90.9091M and sell it for USD 100M.
2. Cover the entire EUR repayment: $90.9091(1.032) =$ EUR 93.8182M.
3. Pay $93.8182(1.122) =$ USD 105.264M under the forward.
4. Add USD 0.150M of fees payable at maturity. All-in cost is **5.414%**, saving USD 0.186M against the direct loan.

Executable Quotes Can Remove the Saving [Core Concept]

Suppose 1.1000 and 1.1220 were indicative mid-market rates. Treasury receives:

$$S_{bid} = 1.0990, \qquad F_{ask} = 1.1230.$$

$$\text{USD repayment} = 100\frac{1.032 \times 1.1230}{1.0990} + 0.150 = 105.6037\text{M}.$$

- Sell borrowed EUR at the **spot bid**. Buy EUR for repayment at the **forward ask**.
- The synthetic route now costs **5.6037%**. Direct USD funding at 5.60% is slightly cheaper.
- The forward already reflects market pricing, including any basis. Adding the same basis again would count it twice.

A Funding Shock Changes Week 5's WACC

[Core Concept]

Week 5 used market-value weights of 50% equity, 35% bonds, and 15% loans:

$$\mathrm{WACC}_0 = 0.50(8\%) + 0.35(6\%) + 0.15(5\%) = 6.85\%.$$

The debt costs above are already after tax. Suppose new bond funding costs **100 bps more before tax**, with a usable 25% tax shield.

$$\Delta\mathrm{WACC} = 0.35(1.00\%)(1 - 0.25) = 0.2625\%, \qquad \mathrm{WACC}_1 = 7.1125\%.$$

- Apply this sensitivity to comparable new projects financed at the assumed target weights.
- Existing fixed-rate debt retains its coupon. A funding shock affects new financing, refinancing, and market values.

Project NPV calculation →

Financing Costs Enter the Valuation Once [Core Concept]

- **Project free cash flow and WACC:** leave interest out of operating free cash flow. Use the relevant after-tax financing cost in WACC.
- **Cash to equity:** include debt proceeds and debt service, then discount at an appropriate cost of equity.
- **Known hedged receipts:** use contracted forward rates for the matched amounts. Uncertain sales volumes and collection dates still require scenarios.
- **A temporary shock:** model the affected financing dates. A one-quarter squeeze does not justify raising every year's hurdle rate permanently.

In-Class Game 8: Project and Funding Committee

[Core Concept]

Role: the US parent's investment committee. Value the Week 5 euro project, then fund it.

- **Stage 1**: Submit ID. Compute the NPV unhedged ($E_0[S_t] = 1.12, 1.14, 1.16$) and hedged ($F_{0,t} = 1.11, 1.12, 1.13$ USD per 1 EUR). Choose an FX policy and Accept or Reject.
- **Stage 2 [Policy shock]**: 20% withholding and a one-year freeze on the Year 2 cash flow, announced before any forward is signed. Recompute the NPV with no usable credit (Case A) and a full credit (Case B).
- **Stage 3 [Funding]**: fund EUR 10M for one year. Local EUR loan at 3.75%, or a parent USD loan at 5% plus an FX swap at mid rates $S = 1.10$, $F = 1.122$ and a EUR 0.05M fee. Compute the all-in EUR cost.
- **Stage 4 [Dealer quotes]**: recompute at the spot ask 1.1020 and forward bid 1.1200. Choose a route and state how financing enters the valuation.
- **Stage 5**: Reflection: *"How did the tax credit change the cost of the withholding tax, and why did the executable quotes change (or not change) your funding choice?"*

Open Week 8 Game

Counts for Week 8 participation.

Game solution →

Micron: Robert Lambert Mobilized Operating Cash [Core Concept]

- **Problem:** manual non-dollar cash forecasts and local funding requests consumed treasury resources. Some currency balances earned negative yields.
- **Action:** Lambert's team implemented automated currency sweeps with Citi into a USD cash pool.
- **Local constraint:** certain entities retained EUR accounts for local tax requirements. The solution swept balances through the UK into the USD pool.
- **Reported benefit:** less manual work and better use of cash. Citi's case reports qualitative benefits, without a complete net savings calculation.

Week 5 connection: internal capital markets depend on accessible cash, currency conversion, and entity-level restrictions.

Cash Pooling Can Reduce External Borrowing [Core Concept]

Illustration. An affiliate holds USD 20M earning 3%. Another borrows USD 20M at 6%. Both positions last 90 days, with ACT/360.

$$\text{Avoidable interest spread} = 20\text{M}(6\% - 3\%)\frac{90}{360} = \$150{,}000.$$

- If transfers are permitted and total fees are USD 20,000, the illustrative net benefit is **USD 130,000** before tax.
- Retain cash needed for local payroll, tax, and stressed operations before sweeping the surplus.
- Across currencies, include conversion and hedging costs. Across entities, verify transfer restrictions and intercompany terms.

A Short Hedge Leaves a Refinancing Decision

[Core Concept]

A five-year USD asset is funded with renewable three-month borrowing and FX hedges. The next year's funding offers are illustrative.

USD 100M funding	Rates over four quarters	One-year interest
One-year locked package	5.0% for the whole year	USD 5.0M
Quarterly renewal, calm	4.5% each quarter	USD 4.5M
Quarterly renewal, stress	4.5%, then 7.0% for three quarters	USD 6.375M

Decision: paying USD 0.5M more in the calm scenario removes three renewal decisions during that year. The five-year asset still needs later funding.

FX-swap maturity evidence →

A Useful Hedge Can Require Cash Today

[Core Concept]

Illustration. A firm will receive USD 10M in six months and sells those dollars forward for EUR.

- If USD strengthens, the future receipt is worth more EUR. The forward position loses value.
- Assume the contract requires collateral and the dealer calls **EUR 0.8M today**. The customer pays only in six months.
- Treasury must fund the EUR 0.8M now. Its final hedged receipt can remain protected while its current liquidity deteriorates.
- The collateral requirement depends on the agreement, thresholds, netting, and eligible assets. Count funding costs and repayment timing in the cash forecast.

Decision: size the hedge and liquidity reserve together.

A Profitable Project Can Miss Tomorrow's Payment [Core Concept]

Illustrative daily cash forecast.

Cash available or due tomorrow	**USD M**
Opening usable cash plus confirmed customer receipts	$20 + 5 = 25$
Committed bank line drawable tomorrow	$+25$
Maturing debt, supplier payments, and collateral call	$-30 - 10 - 12 = -52$
Closing cash before further action	$\mathbf{-2}$

- An extra USD 8M receipt next week makes next week's position positive. Treasury still needs USD 2M by tomorrow.
- Compare usable cash by **currency, entity, and date**. A forecast rollover belongs in a stress scenario until committed.

Why CIP Arbitrage Is Not Frictionless [Core Concept]

Executing a CIP arbitrage trade requires four concurrent legs:

- Borrow USD at an executable wholesale rate (Liability +USD)
- Sell USD for foreign currency in spot market (Spot transaction)
- Lend foreign currency at matched maturity and appropriate credit/collateral terms (Asset +Foreign)
- Enter forward contract to sell foreign currency for USD at maturity (Off-balance sheet forward)

Key Bottleneck:

- Expanding assets and liabilities inflates total balance-sheet size
- Dealer banks face leverage constraints, capital/liquidity requirements, credit and settlement risk, funding spreads, and internal limits. The trade is not literally risk free

Arbitrage T-Accounts →

Dealer Balance-Sheet Constraints

[Core Concept]

Stylized non-risk-based leverage-ratio requirement (Basel III minimum $\bar{X} = 3\%$, Week 7; the US SLR is one implementation):

$$\text{Leverage Ratio} = \frac{\text{Tier 1 Capital}}{\text{Total Exposure (On-Balance Sheet + Off-Balance Sheet)}} \geq \bar{X}$$

- The denominator is not risk weighted, although netting and off-balance-sheet conversion rules still matter. Low-risk positions can therefore consume scarce leverage capacity
- Let $C(B)$ be the total balance-sheet/intermediation cost for position B, so $C'(B)$ is its marginal shadow cost
- Arbitrage condition under balance-sheet limits:

$$|\Delta| \leq C'(B) + \text{other per-unit execution and risk costs}$$

- Arbitrageurs expand the trade only while the basis exceeds the marginal all-in cost. Capacity constraints can leave a nonzero wedge

Stylized dealer model → Hedging-demand evidence →

Convenience Yield of US Dollars

[Core Concept]

The US dollar is the leading reserve, funding, and invoicing currency:

- **Safe-haven and liquidity demand**: global investors value dollar cash, reserves, deposits, and highly liquid dollar securities during stress
- **Convenience premium**: some safe/liquid dollar claims can offer lower yields because they provide transaction, collateral, and liquidity services
- **Impact on basis**: dollar funding demand can contribute to a more negative quoted basis, together with hedging pressure and constrained intermediary balance sheets

Non-US Bank Dollar Funding Gaps

[Core Concept]

Structural mismatch on foreign bank balance sheets:

Dollar Assets

- Corporate loans to global firms
- Trade finance facilities
- US Treasury and agency securities
- Cross-border and project-finance loans

Dollar Funding Sources

- Wholesale funding from money market funds (CDs and repo)
- Commercial paper issuance
- FX swaps that synthesize dollar funding (economically debt-like, though not booked as principal debt in the same way as a deposit)
- Limited retail USD deposits

Retail dollar funding gap = dollar assets minus stable retail dollar deposits. The remainder may be funded with wholesale deposits, bonds, repo, central-bank facilities, or FX swaps. Rollover risk depends on each source's maturity and stability.

Dollar Gap Mechanics → | Funding-gap calculation → | Bank dollar liabilities →

Wholesale Funding Vulnerability

[Core Concept]

Transmission of US wholesale funding shocks to international banks:

- US MMF reform or market panic reduces wholesale USD funding from MMFs
- Non-US banks lose direct USD short-term funding
- Banks scramble to borrow USD synthetically by swapping domestic currency for USD
- A surge in USD demand in the FX swap market can make synthetic funding more expensive and push the quoted basis more negative

Money Market Funds as Dollar Funding Providers [Core Concept]

- MMFs invest in short-term instruments. Government funds emphasize government securities and repos, while prime funds can hold bank certificates of deposit and commercial paper.
- Corporate treasuries and institutional investors use them for cash management. Their yields, liquidity, and credit exposures vary by portfolio and fund structure.
- An MMF share is an investment claim. Stable or floating net asset value and redemption rules depend on the fund. MMF shares carry no deposit insurance.
- Non-US banks can obtain dollar funding from US MMFs through short-term securities and repos. Redemptions can therefore transmit a US investor shock abroad.

MMF instruments and dated reforms →

The 2008 Run on Money Market Funds

[Core Concept]

- After Lehman's failure, losses on its debt caused the Reserve Primary Fund's net asset value to fall below USD 1. Investors redeemed from prime funds.
- Funds reduced purchases of bank and corporate paper. Borrowers lost rollover funding even before their longer-term assets matured.
- Early-redemption incentives and costly asset sales can amplify stress. A stable quoted share price does not imply risk-free underlying assets.
- The US Treasury introduced a temporary guarantee, and Federal Reserve facilities supported short-term funding markets. Reforms subsequently changed liquidity and redemption rules.

Structure of Central Bank Swap Lines [Core Concept]

Bilateral reciprocal currency arrangements between the Fed and foreign central banks:

1. **Leg 1 (reciprocal currency exchange)**: the Fed credits dollars to the ECB and receives an equivalent amount of euros at the prevailing spot rate S. The euros protect the Fed like collateral (Week 7). They are exchanged under the swap rather than pledged separately
2. **Leg 2 (on-lending)**: the ECB supplies dollars to eligible counterparties under its own operation terms and against ECB-eligible collateral
3. **Leg 3 (unwind at maturity)**: the ECB returns dollars plus interest. The Fed returns the euros at the same exchange rate fixed at inception

Risk Allocation:

- The fixed reversal rate removes exchange-rate risk on the principal exchange
- The Fed faces the foreign central bank. The ECB bears the credit risk on its loans to local banks and remains responsible for repayment to the Fed.

Swap Line T-Accounts → Swap-line numerical ledger →

How Swap Lines Backstop Dollar Funding [Core Concept]

Role of central bank swap lines as International Lender of Last Resort (ILOLR):

- Fed sets swap facility lending rate at a fixed benchmark (e.g., OIS + 25 bps)
- Eligible European counterparties can obtain USD from ECB operations at the announced terms and against ECB-eligible collateral
- Facility pricing can create an effective backstop for eligible banks, but stigma, collateral, access, tenor, and counterparty differences prevent a literal universal ceiling
- The facilities can mitigate extreme dollar-funding stress and basis dislocations during crises

2020 bond prices → 2020 bond flows →

Airbus: Securing Liquidity in March 2020 [Core Concept]

Managers: CEO Guillaume Faury and CFO Dominik Asam presented the response on 23 March 2020.

- **Scenario:** COVID-19 disrupted operations and made future cash needs difficult to forecast.
- **Funding action:** management replaced a EUR 5B credit line with a EUR 15B facility, adding **EUR 10B** of capacity.
- **Cash conservation:** Airbus withdrew a dividend proposal worth about EUR 1.4B and suspended voluntary pension top-ups.
- **Analysis:** combine committed funding with lower cash outflows when operating receipts become uncertain.

Airbus: Financing and Investment Adjusted Together [Core Concept]

- **2020 cash pressure:** free cash flow before M&A and customer financing was −EUR 6.935B, including EUR 3.6B of compliance penalties.
- **Longer funding:** Airbus reported issuing bonds to extend financing maturities.
- **Investment response:** project prioritization reduced capital expenditure to about EUR 1.8B, down EUR 0.6B from 2019.
- **Week 5 connection:** capital budgeting must include the cash needed to keep worthwhile projects operating through a disruption.

Discussion: which response buys time, which changes investment, and which retains cash otherwise due to shareholders?

Swap Lines Affect the Bank behind the Company [Core Concept]

- A bank facing expensive dollar funding may raise loan prices or reduce the credit it offers customers.
- A central-bank dollar operation can relieve that funding pressure for eligible banks with acceptable collateral.
- A corporate treasurer still needs an agreed facility, available borrowing capacity, and compliance with draw conditions.
- For a project decision, test whether the firm's own banks can provide the required currency and maturity under stress.

Application: a central-bank announcement is a market signal. The company's signed financing terms determine its access to cash.

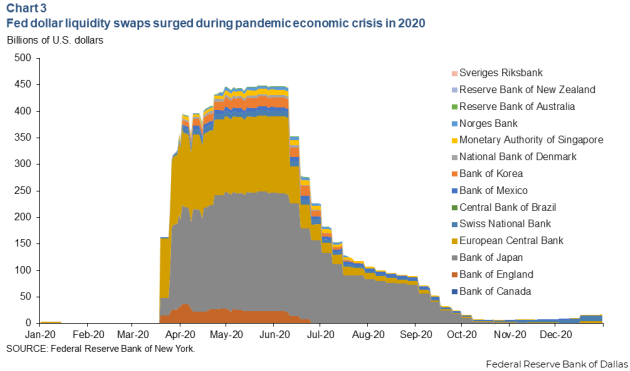
Liquidity Facilities Reach Different Counterparties [Core Concept]

- A central bank can lend its own currency to eligible banks against collateral. Solvency, eligibility, and collateral rules still matter.
- The Federal Reserve's dollar swap lines lend to foreign central banks. The foreign central bank allocates dollars domestically and bears the credit risk on its bank loans.
- MMF and commercial-paper facilities support specific market channels. Asset purchases and Treasury fiscal programs have different counterparties and objectives.
- The 2008 and 2020 interventions combined several tools. Their effects should not all be attributed to swap lines alone.

Earlier international backstops →

History of Crisis Interventions

[Core Concept]



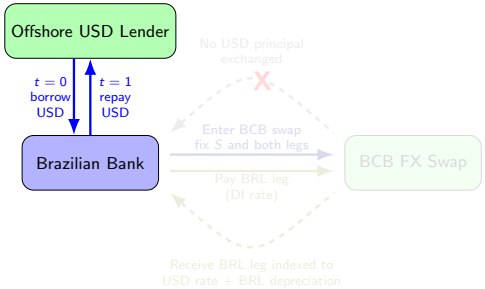
Source: Dallas Fed chart. Underlying data: Federal Reserve Bank of New York

Chronology of Swap Operations → | 1960s dollar-liquidity precedent →

- **2008 GFC**: peak swap line drawings exceeded $580 billion. The Fed had swap lines with 14 central banks
- **2020 Pandemic**: pricing was reduced, weekly 84-day operations were added, and 7-day operations moved temporarily to daily frequency
- **March 2023**: the frequency of 7-day operations moved temporarily from weekly to daily during global banking stress

Brazilian FX Swaps and the Cupom Cambial

[Core Concept]



- The illustrated SCC contract exchanges DI-linked and dollar-linked returns in BRL.
- With spot S and forward F in BRL per USD, the covered implied dollar rate is

$$i_{USD}^{impl} = (1 + i_{BRL})\frac{S}{F} - 1.$$

- Here both interest rates cover the same period. Contract day counts and fixing dates require separate checks.

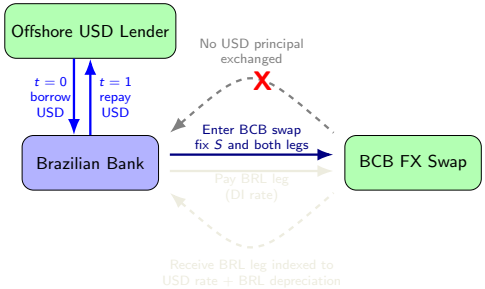
Settlement is a net BRL payment based on the two indexed returns. The bank still needs actual dollars to repay offshore debt.

An NDF settles a net difference in its contractual settlement currency, often USD. Check which currency the hedge actually delivers.

Cupom: numerical steps →

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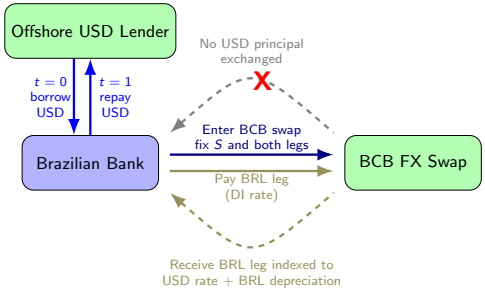
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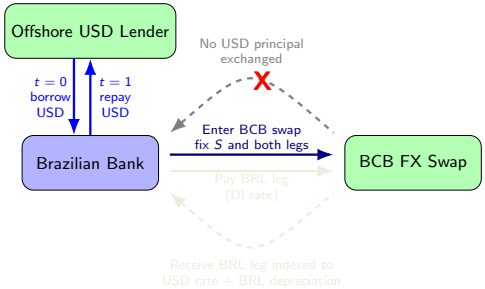
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Cupom: numerical steps →

Deliverability, Market Segmentation, and Basis [Core Concept]

- Deliverable and nondeliverable markets may have different participants, fixing conventions, convertibility constraints, and funding access.
- No physical delivery does not by itself abolish arbitrage. Enforceable cash-settlement links can connect prices, while market restrictions can weaken those links.
- Deliverable swaps can exhibit a persistent basis when balance-sheet capacity is costly. Neither type of forward is automatically an unbiased forecast of future spot.
- Compare onshore and offshore quotes with the same currency units and maturity. A price difference can reflect segmentation and hedge demand as well as expected policy changes.

Debrief: Balance-Sheet Constraints Limit Arbitrage Capacity

Arbitrageurs may not fully eliminate a persistent basis when leverage-ratio requirements, funding spreads, credit risk, and execution costs constrain position size. The wedge can reflect both scarce intermediary capacity and one-sided hedging/funding demand.

Common Misconception Exposed:

- Misconception: treating a non-zero basis as unconstrained free money
- Reality: balance-sheet expansion consumes scarce regulatory leverage capacity

Discussion: *"Why can a funding gap persist when dealers have limited balance-sheet capacity?"*

Example: both forward quotes →

Corporate Treasury Committee: Funding under Stress

[Core Concept]

Your firm needs USD 100M for one year. Use these quotes: direct USD 4%, EUR 2%, spot 1.10, and forward 1.1110.

1. Add USD 0.20M of fees payable at maturity to the synthetic route. Calculate its all-in cost and the saving against direct USD funding.

2. Before execution, the forward rises to 1.1300. Decide which new offer to accept.

3. A customer payment is delayed and the daily cash ladder shows a USD 2M deficit. Propose a funded response with a named source and payment date.

4. State what would change if the forward had already been signed before the market moved.

Committee solution →

Week 9 Connection: The Return after Hedging [Core Concept]

A EUR investor converts EUR 1 into USD at $S = 1.10$, invests at 5%, and hedges the full maturity proceeds at $F = 1.122$ USD per EUR.

$$\text{EUR maturity value} = \frac{1.10(1.05)}{1.122} = 1.029412, \qquad \text{hedged EUR return} = 2.9412\%.$$

- Compare 2.9412% with a EUR investment of similar maturity and risk.
- For a longer bond hedged with short forwards, future renewal prices and collateral needs affect the investment decision.
- A foreign asset's headline yield and its return after hedging answer different questions.

Currency of bank claims → EM currency-market evidence →

What a Junior Treasury Analyst Should Deliver [Core Concept]

1. A comparison of direct and synthetic financing using executable quotes and complete debt service.
2. A valuation that uses the correct cash flows, currency, and financing cost exactly once.
3. A cash forecast by entity, currency, and date, including collateral and failed renewals.
4. A recommendation explaining the cost saving, remaining risk, and funded contingency.

Connection to Week 5: a valuable foreign project also needs a financing plan that can survive its cash-flow risks.

Appendix: Mathematical Proof of NPV Equivalence

[Core Concept]

Let CF_t^* be a foreign-currency cash flow and S_0 be home currency per foreign unit. Suppose the risk-adjusted discount rates and conversion factors satisfy

$$\widehat{S}_t = S_0 \left(\frac{1 + k_h}{1 + k_f} \right)^t .$$

Then the home-currency calculation gives

$$\begin{aligned} \mathrm{NPV}_h &= -S_0 CF_0^* + \sum_{t=1}^{T} \frac{CF_t^* \widehat{S}_t}{(1 + k_h)^t} \\ &= S_0 \left[-CF_0^* + \sum_{t=1}^{T} \frac{CF_t^*}{(1 + k_f)^t} \right] = S_0 \mathrm{NPV}_f . \end{aligned}$$

This equality requires consistent project discount rates and FX conversion. UIP relates interest rates and expected exchange rates; project WACCs also reflect operating and financial risk.

← Back to Main Slide

Example: The Same NPV in Two Currencies

[Core Concept]

A project costs EUR 100 now and pays EUR 110 in one year. Let $S_0 = 1.20$ USD per 1 EUR, $k_f = 10\%$, and $k_h = 5\%$. Assume consistent risk-adjusted conversion.

$$\widehat{S}_1 = 1.20\frac{1.05}{1.10} = 1.14545.$$

$$\mathrm{NPV}_{\mathrm{EUR}} = -100 + \frac{110}{1.10} = 0.$$

$$\mathrm{NPV}_{\mathrm{USD}} = -120 + \frac{110(1.14545)}{1.05} = 0.$$

Note: The two valuations agree under the specified pricing relation. Holding FX at 1.20 while mixing these discount rates would instead give a spurious USD 5.71 NPV.

← Back to Main Slide

Appendix: Step-by-Step 3-Year Capital Budgeting Example [Core Concept]

Project Data: Parent = US firm (USD). Foreign Sub = Eurozone (EUR). Initial CAPEX = EUR 10M ($S_0 = 1.10$ USD per 1 EUR). Home required return $k_h = 10\%$. Terminal recovery is EUR 5M after any disposal tax. No additional CAPEX or working-capital investment is assumed.

Line Item (EUR Millions)	Year 1	Year 2	Year 3
1. Operating Revenue	12.00	15.00	18.00
2. Operating Expenses	(6.00)	(7.50)	(9.00)
3. Depreciation	(2.00)	(2.00)	(2.00)
4. Earnings Before Tax (EBT)	4.00	5.50	7.00
5. Local Corporate Tax (20%)	(0.80)	(1.10)	(1.40)
6. Net Operating Profit After Tax	3.20	4.40	5.60
7. Add Back Depreciation	2.00	2.00	2.00
8. Salvage Value / Recovery	—	—	5.00
9. Free Cash Flow to Sub (EUR)	**5.20**	**6.40**	**12.60**
10. Expected Spot Rate (USD per EUR)	1.12	1.14	1.16
11. Repatriated Cash Flow to Parent (USD)	**5.824**	**7.296**	**14.616**

$$\text{Parent NPV} = -11.000 + \frac{5.824}{1.10} + \frac{7.296}{1.10^2} + \frac{14.616}{1.10^3} = +\mathbf{\$11.306}\text{M}$$

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Example: Subsidiary Cash and Parent Receipts [Core Concept]

A wholly owned affiliate generates EUR 10M of after-tax free cash flow in Year 1. It retains EUR 4M under a remittance limit and distributes EUR 6M.

Step	Amount
Dividend before withholding	EUR 6.00M
Withholding at 10%, with no usable credit	EUR 0.60M
Cash reaching parent	EUR 5.40M
Convert at 1.10 USD per EUR	USD 5.94M

The EUR 4M retained belongs in a separate forecast of local investment returns, later distributions, or terminal value.

Interpretation

This year's parent receipt differs from subsidiary free cash flow. Giving the retained funds zero value would omit an asset still owned by the group.

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Appendix: Tax Credit Limit & Blocked Funds Calculation

[Core Concept]

Case 1: Simplified Foreign Tax Credit Ceiling

Suppose a home jurisdiction uses a pure worldwide credit system. Foreign-source taxable income is EUR 10M, qualifying foreign tax is 25%, and the home tax rate is 21% (ignoring baskets, participation exemptions, minimum taxes, and expense allocation).

- Local Tax Paid = EUR 2.5M.
- Tentative Home Tax before credit = EUR 10M $\times$ 21% = EUR 2.1M.
- Usable FTC = $\min(2.5\text{M}, 2.1\text{M})$ = EUR 2.1M. Current residual home tax is zero. Use of the EUR 0.4M excess in other periods depends on domestic carryover rules.

Case 2: Blocked Funds Discounting

Suppose Year 1 cash flow of EUR 5.2M is blocked by host government for 2 years and earns only 2% local bank interest, while parent cost of capital is 10%.

- Accumulated Cash in Year 3 = EUR 5.2M $\times (1.02)^2$ = EUR 5.41M.
- PV at Parent in Year 0 = $\frac{5.41\text{M} \cdot E_0[S_3]}{(1+k_h)^3}$, creating a significant value reduction vs immediate remittance.

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Example: The Cost of a Two-Year Remittance Delay [Core Concept]

At the end of Year 1, EUR 5M is available for distribution. Assume constant FX, a 10% required return, no extra tax, and a certain release after two years.

Use of funds	Value at the end of Year 1
Remit immediately	EUR 5.000M
Retain at 2%, release in Year 3	$5(1.02)^2/(1.10)^2 =$ EUR 4.299M
Retain at a comparable-risk 10% return	$5(1.10)^2/(1.10)^2 =$ EUR 5.000M

Interpretation

The value loss is EUR 0.701M in the 2% case. A restriction has a smaller cost when local funds earn an appropriate risk-adjusted return.

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Example: A Transfer Price That Blocks a Good Sale

[Core Concept]

A component division has spare capacity and marginal cost USD 60. An assembly division can finish the product for USD 20 and sell it outside the group for USD 100.

Internal transfer price	Assembly division margin	Group margin if produced
USD 70	$100 - 70 - 20 = 10$	$100 - 60 - 20 = 20$
USD 90	$100 - 90 - 20 = -10$	$100 - 60 - 20 = 20$

If the assembly manager can refuse a loss-making order, a USD 90 internal price can prevent production.

Interpretation

The group loses USD 20 of contribution despite a profitable outside sale.

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Appendix: Historical Profit Shifting and Loan Taxation [Core Concept]

Stylized pre-2017 tax-rate exercise, with USD millions. Initial US profit is 400 and Irish profit is 50. Assume US tax 35%, Irish tax 12.5%, and no residual taxes.

Scenario	US tax	Irish tax	Total tax
Baseline	140.00	6.25	146.25
Shift 150 of royalties to Ireland	87.50	25.00	112.50
Also deduct Irish interest of 50	87.50	18.75	106.25

- Royalty shifting saves 33.75. The extra deduction saves at most 6.25 only if the recipient's interest income bears no tax. Combined saving is then 40.00.
- If the lender pays tax on the interest, subtract that tax from the saving. The baseline must retain the same real Irish profit of 50.
- This exercise uses historical tax assumptions. Arm's-length pricing, interest limitations, anti-hybrid rules, and residual taxes can eliminate the apparent benefit.

Source: OECD Transfer Pricing Guidelines (2022).

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Appendix: Historical Royalty-Shifting Arithmetic [Core Concept]

Hypothetical pre-2017 tax exercise, USD millions. An Irish affiliate pays a US parent for intellectual property. Assume 35% US tax, 12.5% Irish tax, and no residual home tax.

- Benchmark royalty: 200. Reduced royalty: 50. Paying 150 less raises Irish profit by 150 and lowers US profit by 150.
- US profit falls from 400 to 250. Irish profit rises from 50 to 200. Group pretax profit remains 450.
- Total tax falls from $400(0.35) + 50(0.125) = 146.25$ to $250(0.35) + 200(0.125) = 112.50$.
- The arithmetic saving is $150(0.35 - 0.125) = 33.75$.

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Example: Reversing the Direction of the Internal Sale

[Core Concept]

Use hypothetical tax rates: parent 20%, subsidiary 30%. This time the parent produces for EUR 80 and the subsidiary resells outside the group for EUR 200.

Transfer price	Parent profit	Subsidiary profit	Combined tax
EUR 100	20	100	$0.20(20) + 0.30(100) = 34$
EUR 140	60	60	$0.20(60) + 0.30(60) = 30$

Interpretation

Raising the transfer price now shifts profit to the lower-tax parent and reduces tax by EUR 4. The direction reverses the main slide's result.

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Example: Profit, Depreciation, and Free Cash Flow

[Core Concept]

A project has revenue 12, cash operating costs 6, depreciation 2, tax rate 20%, new CAPEX 1, and an increase in working capital of 0.5. All amounts are EUR M.

Step	Calculation	EUR M
EBIT	$12 - 6 - 2$	4.0
Operating profit after tax	$4(1 - 0.20)$	3.2
Operating cash flow	$3.2 + 2$	5.2
Free cash flow	$5.2 - 1 - 0.5$	3.7

Interpretation

Depreciation reduces taxes by EUR 0.4M. Deduct cash spent on equipment and working capital from operating cash flow to obtain FCF.

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Appendix: Three-Year Project, Year 1

[Core Concept]

All amounts are millions. The exchange rate is 1.10 USD per 1 EUR and the USD discount rate is 10%.

Step	Amount
Revenue less cash costs	30 - 18 = 12 EUR
Less depreciation	12 - 5 = 7 EUR
Host tax at 25%	1.75 EUR
Operating cash after tax	$5.25 + 5 = 10.25$ EUR
Cash after 10% withholding	$10.25(0.90) = 9.225$ EUR
Convert to USD	$9.225(1.10) = 10.1475$ USD
Discount to date 0	$10.1475/1.10^1 = 9.2250$ USD

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Appendix: Three-Year Project, Year 2

[Core Concept]

All amounts are millions. The exchange rate is 1.10 USD per 1 EUR and the USD discount rate is 10%.

Step	Amount
Revenue less cash costs	32 - 19 = 13 EUR
Less depreciation	13 - 5 = 8 EUR
Host tax at 25%	2.00 EUR
Operating cash after tax	6.00 + 5 = 11.00 EUR
Cash after 10% withholding	11.00(0.90) = 9.900 EUR
Convert to USD	9.900(1.10) = 10.8900 USD
Discount to date 0	$10.8900/1.10^2 = 9.0000$ USD

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Appendix: Three-Year Project, Year 3

[Core Concept]

All amounts are millions. The exchange rate is 1.10 USD per 1 EUR and the USD discount rate is 10%.

Step	Amount
Revenue less cash costs	34 - 20 = 14 EUR
Less depreciation	14 - 5 = 9 EUR
Host tax at 25%	2.25 EUR
Operating cash after tax	6.75 + 5 = 11.75 EUR
Add salvage (equals tax book value)	12.00 EUR
Cash after 10% withholding	23.75(0.90) = 21.375 EUR
Convert to USD	21.375(1.10) = 23.5125 USD
Discount to date 0	$23.5125/1.10^3 = 17.6653$ USD

The exercise applies withholding to all terminal cash. If sale proceeds differ from tax book value, include tax on the gain or loss before remittance.

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Appendix: Transfer-Price Markup, Both Entities

[Core Concept]

EUR millions in Year 1	Baseline	Additional parent markup of 4
Subsidiary revenue	30.00	30.00
Subsidiary cash costs	18.00	22.00
Depreciation	5.00	5.00
Taxable profit	7.00	3.00
Host tax at 25%	1.75	0.75
Cash before withholding	10.25	7.25
Dividend after 10% withholding	9.225	6.525
Additional parent income after 20% tax	0	3.200
Combined parent receipts	9.225	9.725

Consolidated receipts increase by EUR 0.5M.

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Appendix: Reading Multinational Financial Statements [Core Concept]

- Market capitalization equals share price times shares outstanding on a specified date. Revenue and net income measure flows over a reporting period.
- For valuation, distinguish equity value from enterprise value. A common enterprise-value measure adds debt and subtracts excess cash, with adjustments for other claims.
- Comparisons among technology, energy, and manufacturing MNCs require common periods, currencies, and accounting definitions. Use each profit measure consistently: operating income, EBITDA, and net income deduct different expenses.
- Geographic revenue can be classified by customer location. Production location, incorporation, and taxable income require separate information.

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Appendix: Revenue, Profit, and Location — Two MNC Examples [Core Concept]

Amounts are USD billions, using each company's fiscal 2025 GAAP results.

Company	Global revenue	Net income	Fiscal year ended
NVIDIA	130.497	72.880	January 26, 2025
Microsoft	281.724	101.832	June 30, 2025

- Microsoft's revenue by customer location was USD 144.546B in the US and USD 137.178B elsewhere. These figures classify sales by customer location.
- Net income differs from operating income, EBITDA, and operating cash flow. Fiscal years also end on different dates.
- Calculating market capitalization requires share price and shares outstanding at a specified date.

Sources: NVIDIA FY2025 results; Microsoft FY2025 annual report.

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Appendix: Comparing Multinational Companies [Core Concept]

These multinational firms span different businesses and asset requirements.

Examples	Valuation questions
NVIDIA, Broadcom, TSMC	IP ownership, fabrication, customers, and geographic production
Microsoft, Apple, Alphabet, Meta	Intangibles, platforms, and the location of sales versus taxable income
Amazon, Tesla	Logistics or production capacity, working capital, and margins
Saudi Aramco	Commodity prices, physical reserves, and capital expenditure
Berkshire Hathaway	Different operating businesses, investments, and financial claims

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Example: A Guarantee Hidden in an Affiliate

[Core Concept]

A parent approves a foreign affiliate's factory with NPV USD 3M. A local manager also guarantees USD 8M of a supplier's debt without central review.

- Assume a 25% probability of full payment on the guarantee next year, zero recovery, and a 10% risk-adjusted discount rate.
- The guarantee's expected payment is $0.25(8) = \text{USD } 2\text{M}$, with PV $2/1.10 = \text{USD } 1.82\text{M}$.
- Net value after this additional obligation is $3 - 1.82 = \text{USD } 1.18\text{M}$. The board needs entity-level reporting and approval limits to see the exposure.

Interpretation

Governance affects value by changing actual obligations and investment choices.

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Example: A controller transfers value to another firm [Core Concept]

A controller owns 60% of Firm A and all of Firm B. Firm A sells an asset worth 10 to Firm B for 6. Use a common currency unit and ignore tax.

Owner / entity	Value effect
Firm A	-4.0
Controller through its stake in A	$0.60(-4) = -2.4$
Controller through B	$+4.0$
Controller net gain	$+1.6$
Minority shareholders of A	$0.40(-4) = -1.6$

Interpretation

The controller gains exactly what A's minority owners lose. Independent valuation and approval by disinterested shareholders can constrain this conflict.

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Appendix: Renault–Nissan and Control Rights [Core Concept]

- The historical Renault–Nissan alliance combined cross-shareholdings, shared management, and state influence through the French government's stake in Renault.
- Cash-flow ownership and voting control can differ from each partner's economic contribution. A partner can perceive its influence as too small relative to its sales or technology contribution.
- Joint decisions about leadership, investment, and electric vehicles can therefore create conflicts among partners and their shareholders.
- **Discussion**: which decisions belong to each company's board, and which require agreement across the alliance? What protections would minority investors want?

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Appendix: Enron and Opaque Group Structures

[Core Concept]

- Enron's use of complex special-purpose entities and related-party arrangements obscured debt, risks, and the quality of reported earnings.
- The governance problem was misleading reporting and conflicts of interest. Evaluate these issues within each entity.
- Creditors and shareholders need to identify guarantees, related-party exposures, and the economic substance of transactions across the group.
- **Discussion**: which reported profits would you reconcile to cash, and what obligations might sit outside the apparent operating balance sheet?

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Appendix: Samsung and Minority-Shareholder Transfers [Core Concept]

- The Samsung SDS example discussed in Bae, Kang, and Kim (2002) concerns securities issued to members of a controlling family on terms criticized as favorable to the recipients.
- The historical case contrasts a subscription price of KRW 7,150 with an indicated outside-market price of KRW 55,000 for the relevant share entitlement.
- The economic question is whether the transaction transfers value away from outside shareholders. Valuation date, security terms, and allocation rights must be comparable.
- Pyramids and cross-holdings can separate control from cash-flow ownership. Independent review and minority approval can constrain abusive transactions.

Source: Bae, Kang, and Kim (2002), historical case.

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Appendix: Vale and Monitoring from a Distance

[Core Concept]

- The Vale example asks how a foreign investor can monitor a geographically dispersed business governed by Brazilian institutions.
- Physical distance, language, operational complexity, and local enforcement can raise the cost of evaluating management and coordinating shareholders.
- Assess Vale using its reporting framework, filings, and Brazilian enforcement institutions.
- **Discussion**: how would local expertise, site-level disclosure, independent directors, and external assurance change your willingness to provide capital?

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Appendix: Volkswagen, Boards, and Enforcement [Core Concept]

- A two-tier system separates management from a supervisory board. Representation by major shareholders, employees, or public bodies can create competing objectives.
- Volkswagen's emissions scandal illustrates that formal oversight structures can fail to prevent misconduct. The case raises questions about board information, incentives, and enforcement.
- In January 2017, Volkswagen agreed to plead guilty and pay USD 4.3B in US criminal and civil penalties. Individual executives and employees also faced charges.
- **Valuation link**: misconduct can create fines, remediation costs, lost sales, and financing constraints across multiple jurisdictions.

Source: US Department of Justice, January 11, 2017.

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Appendix: Petrobras ADRs and the Limits of Cross-Listing [Core Concept]

- An ADR allows investors to hold an interest in a foreign issuer's shares through a depositary arrangement. Exchange-listed programs bring disclosure and regulatory obligations.
- Petrobras's access to US investors exposed its disclosures to US securities enforcement. In 2018, the SEC announced a settlement over misleading investors.
- This illustrates both the potential value and the limits of regulatory bonding. Additional oversight can sanction wrongdoing without preventing every abuse.
- **Discussion**: how could changes in disclosure and enforcement affect investors' required return after a cross-listing?

Source: SEC, Petrobras settlement, September 27, 2018.

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Appendix: Legal Origins and Investor Protection [Core Concept]

- La Porta, Lopez-de-Silanes, Shleifer, and Vishny compared investor protections and ownership concentration across countries with different legal origins.
- Their historical findings associate common-law systems with stronger measured shareholder protection than some civil-law groups, including the French-origin group.
- The mechanism is financing: outside investors may supply less capital or demand control when they expect expropriation and weak remedies.
- The result is an empirical association for particular samples and measures. Reforms, enforcement, ownership, and issuer-specific arrangements matter for a current investment decision.

Source: La Porta et al., Law and Finance.

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Appendix: Siemens and Cross-Border Enforcement [Core Concept]

- In 2008, coordinated US and German actions resulted in approximately USD 1.6B of penalties against Siemens for bribery-related misconduct.
- Cross-border business creates exposure to more than one enforcement system. A transaction may need to satisfy rules in several jurisdictions.
- Boards and managers need controls over agents, payments, approvals, and financial reporting. External audits and credible sanctions complement internal monitoring.
- **Valuation link**: expected penalties and disruption can reduce cash flows. Assess any financing-cost effect using separate evidence.

Source: US Department of Justice, December 15, 2008.

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Example: A Foreign Project Committee Decision [Core Concept]

A firm compares two one-year foreign-market strategies. All figures are USD M, FX is fixed, and the risk-adjusted required return is 10%.

Mode	Investment today	Year 1 parent cash	NPV
Export	4.0	5.0	$5/1.10 - 4 = 0.545$
Owned affiliate	8.0	10.0	$10/1.10 - 8 = 1.091$

A proposed guarantee adds USD 0.7M in present-value costs to the affiliate. Its revised NPV is $1.091 - 0.7 = 0.391$, below exporting.

Interpretation

Compare parent NPVs after investment, financing obligations, and governance costs for each entry mode.

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Appendix: CIP Mathematical Derivation and Arbitrage Bounds [Core Concept]

No-Arbitrage Proof: Consider two strategies for investing 1 unit of home currency H (the QUOTE currency):

1. **Strategy A (Domestic Investment)**: invest 1 unit of H at the home rate i_H. Yields $1 + i_H t$ at time t.

2. **Strategy B (Covered Foreign Investment)**:
   - Convert 1 unit of QUOTE to BASE at spot rate S (BASE/QUOTE): yields $1/S$ units of BASE
   - Invest $1/S$ units of BASE at the foreign rate i_F: yields $(1/S)(1 + i_F t)$ units of BASE
   - Sell BASE forward at forward rate F (BASE/QUOTE): yields $(F/S)(1 + i_F t)$ units of QUOTE

By no-arbitrage, Strategy A return equals Strategy B return:

$$1 + i_H t = \frac{F}{S}(1 + i_F t) \implies F^* = S \times \frac{1 + i_H t}{1 + i_F t}$$

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Appendix: What the CIP Evidence Establishes

[Core Concept]

- Cerutti, Obstfeld, and Zhou document persistent post-crisis covered-interest-parity deviations across major currencies. This is not a claim that every currency has the same basis.
- Their 2019 IMF working-paper version uses a baseline weekly panel of eight currencies over 2002–2018. Sample frequency and dates should be tied to the version being discussed.
- Dollar strength, risk sentiment, funding conditions, and policy and regulatory changes help explain variation. Their importance differs across currencies and periods.
- A quoted wedge is not a free arbitrage profit: actual borrowing rates, credit exposure, margin, settlement, and balance-sheet costs must enter the trade.

Source: Cerutti, Obstfeld and Zhou, IMF WP/19/14.

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Appendix: T-Account Expansion of CIP Arbitrage Trade [Core Concept]

T-account entries for a USD cash borrower lending in EUR with FX swap cover:

Arbitrageur Balance Sheet (Time 0)

- **Assets**:
  - EUR Interbank Loan: +$100M equivalent
- **Liabilities**:
  - USD Wholesale Borrowing: +$100M
- **Off-Balance Sheet**:
  - Forward Contract (Sell EUR / Buy USD)

Regulatory Impact

- Gross assets increase by $100M
- Gross liabilities increase by $100M
- Leverage ratio denominator inflates by $100M
- Is largely FX-hedged but still carries funding, credit, settlement, liquidity, and model/basis risks

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Appendix: A Constrained Dealer Chooses Its Position

[Core Concept]

Assume a positive annualized wedge Δ, quadratic funding/balance-sheet cost, and a position limit $\bar{B}$:

$$\max_{0 \le B \le \bar{B}} \left\{ \Delta B - \frac{\gamma}{2} B^2 \right\}, \qquad B^* = \min\left(\frac{\Delta}{\gamma}, \bar{B} \right), \quad \gamma > 0.$$

1. Without a binding limit, marginal revenue equals marginal cost: $\Delta = \gamma B^*$.
2. At a binding limit, $\Delta = \gamma \bar{B} + \mu$, where $\mu \ge 0$ is the capacity shadow price. Equality with $\gamma \bar{B}$ alone need not hold.
3. Example: $\Delta = 0.006$, $\gamma = 0.0001$ per million, and $\bar{B} = 40$ million. Desired position is 60 million, but the dealer supplies 40. $\mu = 0.002$ (20 bps).

This is a stylized supply model, not a structural estimate of the basis.

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Appendix: Non-US Bank Dollar Funding Gap Mechanics [Core Concept]

Interpreting the Non-US Bank Dollar Funding Gap:

- Non-US banks hold trillions of dollars of loans and securities, but typically have a smaller base of stable retail dollar deposits
- The resulting *retail funding gap* is not an accounting shortfall: it is funded through a changing mix of:
  - FX swaps and other currency-hedging instruments
  - Interbank deposits, repo, and secured funding
  - Commercial paper, certificates of deposit, money-market funds, and longer-term debt
- Vulnerability depends on maturity, collateral, counterparty concentration, and access to central-bank liquidity. Simple headline totals are not additive across datasets.

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Appendix: Federal Reserve Foreign-Currency Swap T-Account Mechanics [Core Concept]

T-account entries for a $1 billion Fed-ECB swap line draw:

Federal Reserve Balance Sheet

- **Assets**:
  - Central Bank Liquidity Swap claim on ECB: +$1B equivalent
- **Liabilities**:
  - ECB Deposit Account (USD Cash): +$1B

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European Central Bank (ECB)

- **Assets**:
  - Fed Deposit Account (USD Cash): +$1B
- **Liabilities**:
  - Euro deposit liability to the Fed under the swap: +$1B equivalent

Appendix: Chronology of Federal Reserve Swap Line Interventions [Core Concept]

Historical Timeline of Global Swap Operations:

- **December 2007**: Fed establishes initial temporary swap lines with ECB and SNB
- **October 2008**: selected major central-bank lines moved to full-allotment operations, while the network expanded substantially
- **May 2010**: Re-activation during European sovereign debt crisis
- **October 2013**: Fed converts C5 swap arrangements into permanent standing facilities
- **March 2020**: pricing was reduced by 25 bps, weekly 84-day operations were added, and 7-day operations moved temporarily to daily frequency
- **March 2023**: 7-day operations moved temporarily from weekly to daily frequency during global banking stress

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FX Swap: Match the Two Dates Explicitly

[Core Concept]

A firm must pay EUR 1M on day 10 and expects to receive EUR 1M on day 20. It arranges an FX swap with USD per EUR rates 1.1010 and 1.1020 for those dates.

	Day 10	Day 20
Commercial flows	Pay EUR 1M	Receive EUR 1M
FX-swap flows	Receive EUR 1M. Pay USD 1.101M	Pay EUR 1M. Receive USD 1.102M
Combined flows	Pay USD 1.101M	Receive USD 1.102M

1. The EUR cash flows cancel on both dates, provided the customer's receipt arrives as expected.
2. The firm still needs USD funding for ten days. The $1,000 difference is a swap-point amount, not profit after funding costs.

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Basis Signs: Keep the Exchange-Rate Units Fixed

[Core Concept]

Use the main slide's quote: S, F in USD per foreign unit, rates per year, horizon t. Suppose $S = 1.10$, $F = 1.122$, $i_F = 2\%$, $i_{USD} = 3\%$, and $t = 1$.

1. To raise USD 1, borrow $1/S = 0.909091$ foreign units.
2. Foreign repayment is $(1/S)(1.02) = 0.927273$. Covered dollar repayment is $0.927273(1.122) = 1.0404$.
3. Synthetic dollar rate is 4.04%. $\Delta = 4.04\% - 3.00\% = 1.04\% = 104$ bps.
4. Foreign-leg quoted basis is $b = -\Delta S/F = -0.0104(1.10/1.122) = -1.0196\%$, or -101.96 bps.

If using inverse quotes $S_i = 1/S$ and $F_i = 1/F$ (foreign units per USD), the repayment is $(1 + i_F t)S_i/F_i$. The economics is unchanged, but the ratio of forward to spot is inverted.

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Synthetic Borrowing, Step 1: Principal and Interest [Core Concept]

One-year illustration: EUR borrowing rate 5%, USD rate 10%, spot $S = 1.10$ USD per EUR. The frictionless forward is

$$F = 1.10\frac{1.10}{1.05} = 1.15238095 \text{ USD per EUR.}$$

	Today	In one year
EUR loan	Borrow EUR 10,000	Repay EUR 10,500
Spot conversion	Sell EUR 10,000. Receive USD 11,000	—
Forward hedge	Contract to buy EUR 10,500	Pay $10{,}500 \times F = \$12{,}100$

Net financing: receive \$11,000 today and repay \$12,100 in one year. Cost $= 12{,}100/11{,}000 - 1 = 10\%$. Hedge the *interest as well as the principal*.

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Synthetic Borrowing, Step 2: Borrower-Specific Spreads [Core Concept]

Use the same market forward, but suppose this particular firm can borrow EUR at 8% or USD at 16%.

1. Borrow EUR 10,000 and convert at 1.10 to receive USD 11,000 today.
2. EUR repayment is $10{,}000 \times 1.08 = 10{,}800$.
3. Forward-cover that repayment: $10{,}800 \times 1.15238095 = \$12{,}445.71$.
4. Synthetic dollar cost is $12{,}445.71/11{,}000 - 1 = 13.1429\%$.
5. Direct dollar repayment would be $11{,}000 \times 1.16 = \$12{,}760$. The synthetic route saves \$314.29 before transaction costs.

Market CIP can hold while a firm's funding alternatives differ: its credit spreads need not be equal across markets.

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Convert a Basis-Point Quote into a Dollar Cost [Core Concept]

Illustration using a 47 bp annualized wedge, USD 100M principal, and a three-month period treated as $\tau = 0.25$.

1. Convert basis points: 47 bps $= 47/10{,}000 = 0.0047$ per year.
2. Prorate the spread: $0.0047 \times 0.25 = 0.001175$, or 0.1175% for the quarter.
3. Multiply by principal: $100{,}000{,}000 \times 0.001175 = \$117{,}500$ extra financing cost.
4. At a 386 bp annualized wedge, the same calculation gives $100\text{M} \times 0.0386 \times 0.25 = \$965{,}000$.

The arithmetic treats each quoted wedge as constant over the contract period. A peak daily observation is not a realized average cost for every borrower.

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Money-Market Funds: Cash Management and Run Protection

[Core Concept]

- Investors buy fund shares. Funds hold short-term instruments such as Treasury bills, repo, bank CDs, and commercial paper, depending on the fund type.
- A fund share is not an insured bank deposit. Some holdings can qualify as cash equivalents for accounting purposes when the relevant criteria are met. That does not make them riskless.
- **2014 US reform:** institutional prime and tax-exempt funds moved to floating NAV. Reforms also introduced liquidity-fee and redemption-gate provisions.
- **2023 US reform:** increased liquidity requirements, removed the redemption-gate mechanism, and revised liquidity fees. The 2014 design should not be taught as an unchanged current rule.

Source: SEC, 2014 reforms.

Source: SEC, 2023 reforms.

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A Funding Gap Is a Maturity Problem, Not Missing Accounting [Core Concept]

Illustrative non-US bank: USD 1,000M assets financed by 400M retail deposits, 300M FX swaps, 200M wholesale CDs, and 100M equity/long-term debt.

1. Total funding equals assets: $400 + 300 + 200 + 100 = 1{,}000$.
2. The gap relative to retail deposits is $1{,}000 - 400 = 600$. Other funding fills it. There is no unbalanced balance sheet.
3. Tomorrow, 150M of swaps and 100M of CDs mature. Available liquid dollars are 80M and contractual inflows are 20M.
4. With no rollover, the cash shortfall is $150 + 100 - 80 - 20 = 150$M. The bank must obtain dollars, sell assets, or reduce lending.

Diversifying maturities and counterparties can reduce rollover exposure even if the headline 600M retail funding gap is unchanged.

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Central-Bank Swap Line: Cash Flows and Risk Allocation [Core Concept]

Assume a USD 1 billion, seven-day swap at EUR/USD = 1.10 USD per EUR and a 4.25% dollar rate, ACT/360. These are illustrative terms.

1. **Opening exchange:** the Fed provides USD 1B to the ECB. The ECB provides $1B/1.10 = \text{EUR } 909.09M$.
2. **Local operation:** the ECB lends the dollars to eligible banks against collateral under its own operation's terms.
3. **Interest:** $1B \times 0.0425 \times 7/360 = \$826{,}388.89$.
4. **Closing exchange:** the ECB returns USD 1B plus dollar interest. The Fed returns the original EUR principal at the original exchange rate.

The fixed reversal rate removes exchange-rate exposure on the principal exchange. The foreign central bank bears credit risk on its local bank lending and remains responsible to the Fed.

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Cupom Cambial: Compute the Implied Dollar Rate

[Core Concept]

One-year example: spot $S = 5.00$ BRL per USD, forward $F = 5.25$ BRL per USD, domestic BRL interest rate 12%.

1. Start with USD 1, sell it spot, and invest BRL 5.00.
2. BRL maturity value is $5.00 \times 1.12 = 5.60$.
3. Buy dollars forward at 5.25 BRL per USD: $5.60/5.25 = \$1.066667$.
4. The covered implied dollar rate is $S(1 + i_{BRL})/F - 1 = 6.6667\%$.

The approximate calculation $12\% - 5\% = 7\%$ ignores compounding. Brazilian central-bank FX-swap contracts exchange indexed returns in local currency. A hedge can provide dollar-price exposure without delivering dollar funding offshore.

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Example: Calculate Both Routes before Choosing [Core Concept]

Raise USD 100M for one year. Direct USD rate 4%. EUR rate 2%. Spot 1.10 USD per EUR. Ignore fees and collateral costs.

1. Borrow $100/1.10 =$ EUR 90.9091M. Exchange for USD 100M. EUR repayment is $90.9091 \times 1.02 = 92.7273$M.

2. At forward $F = 1.111$ USD per EUR, repay $92.7273 \times 1.111 =$ \$103.02M. Synthetic rate is 3.02%; direct repayment is \$104M. Saving: \$0.98M.

3. After the quoted forward rises to 1.13, a *new* trade would repay $92.7273 \times 1.13 =$ \$104.7818M, a 4.7818% rate.

4. For a new borrowing decision, direct USD funding now saves \$0.7818M. A previously locked forward retains its contracted rate. It is not repriced retroactively.

Debrief: distinguish the funding cost of new trades from the mark-to-market value and collateral needs of existing trades.

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Offshore Dollar Backstops before 2008

[Core Concept]

- Offshore dollar banking expanded faster than an explicit international safety net. There was no single offshore dollar central bank, but that does not mean official support was always absent.
- In 1974, Franklin National's foreign-exchange losses and wholesale funding problems led to substantial Federal Reserve discount-window support. Authorities worried about contagion through the Eurodollar market.
- Central-bank discussions at the BIS addressed cross-border liquidity and supervisory responsibilities after the 1974 bank failures. Home and host authorities had to coordinate their roles.
- The 1980s Latin American debt crisis created a different problem: official bridging finance and bank-debt negotiations addressed sovereign repayment difficulties and bank exposures, not just a temporary FX-swap-market squeeze.

Source: Federal Reserve Bank of Cleveland, historical account.

Source: Federal Reserve, July 1974 BIS meeting memorandum.

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The 1960s Swap Network Already Addressed Offshore Liquidity [Core Concept]

- The Federal Reserve's swap network began in 1962 and continued through the transition from fixed to floating exchange rates.
- McCauley and Schenk's archival research shows the Fed, BIS, and European central banks used swaps to manage Eurodollar funding conditions, including seasonal dollar shortages and interest-rate spikes.
- These operations had domestic-policy motives as well as international ones: offshore funding strains could interfere with US monetary-policy transmission.
- The later 2008 and 2020 operations greatly expanded dollar liquidity provision. Their use for offshore funding was an extension of an older function, not an entirely new purpose invented in 2008.

Source: McCauley and Schenk, BIS Working Paper 851, 2020.

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Newell: Questions behind the Reported Saving [Core Concept]

- Which EUR cash flows or net investment support the EUR obligation? How much currency risk remains at maturity?
- What are the swap's coupon dates, final principal exchange, collateral terms, and early termination cost?
- What is the comparable direct EUR funding offer after fees and tax?
- The reported 200 bp benefit combines the actual transaction's economics. The interview does not identify how much came from the basis rather than interest-rate or credit-spread differences.

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Project NPV under the Financing Sensitivity

[Core Concept]

Assume an initial USD 100M investment and USD 24.3M of annual free cash flow for five years.

$$\mathrm{NPV}(k) = -100 + 24.3 \sum_{t=1}^{5} \frac{1}{(1+k)^t}.$$

	WACC	NPV, USD M
Week 5 financing benchmark	6.8500%	+0.0362
100 bp increase on the bond component	7.1125%	−0.6646

This marginal project changes sign under the assumed persistent funding shock. Reassess operating forecasts and feasible financing before deciding.

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Committee Solution and Decision Conditions

[Core Concept]

1. Initial synthetic repayment: $100(1.02)(1.111/1.10) + 0.20 = 103.22$. Cost: **3.22%**. Saving against direct USD funding: **USD 0.78M**.

2. With $F = 1.13$, new synthetic repayment is **USD 104.9818M**. Direct funding at 4% saves **USD 0.9818M**.

3. The USD 2M cash deficit requires additional usable cash by the due date: for example, a confirmed draw or a permitted internal transfer. A later receipt alone does not meet the payment.

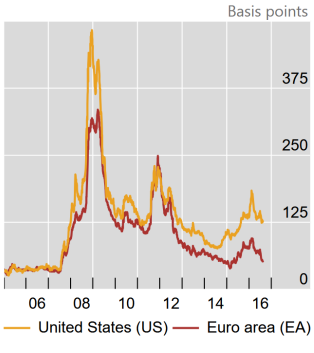
4. A signed forward keeps its contractual rate. The market move can change its value, collateral needs, and replacement cost.

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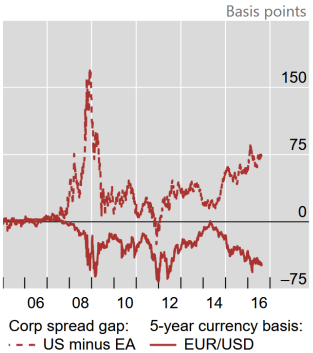
Firms Use FX Swaps When Borrowing in Foreign Currencies

[Core Concept]

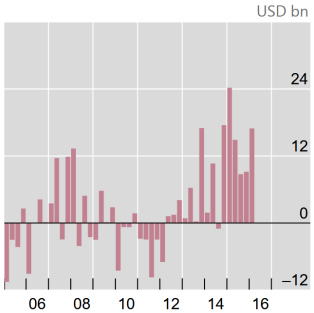
Corporate asset swap spread



Spread differential and the basis



US non-financial firms' EUR debt issuance



Source: BIS.

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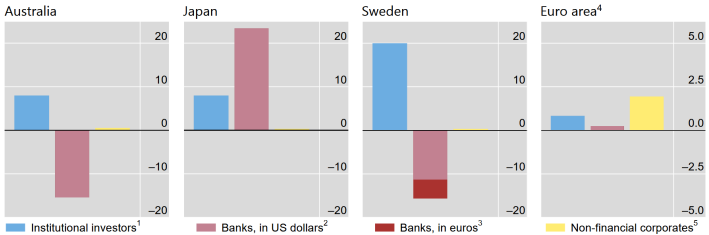
CIP Deviations Follow Banks' FX Swap Demand to Hedge Lending

[Core Concept]

Currency hedging by banks, institutional investors and non-financial corporates

As a percentage of 2015 GDP

Graph 2



[1] Foreign currency securities holdings of institutional investors (eg pension funds and insurance companies) multiplied by the respective currency hedge ratios; for the euro area, US dollar debt securities holding only, assuming 100% currency hedge ratios. [2] Each jurisdiction's BIS reporting banks' consolidated net US dollar assets. [3] Each jurisdiction's BIS reporting banks' consolidated net euro assets. [4] 2015 quarterly average. [5] Local currency debt outstanding issued by US non-financial corporations.

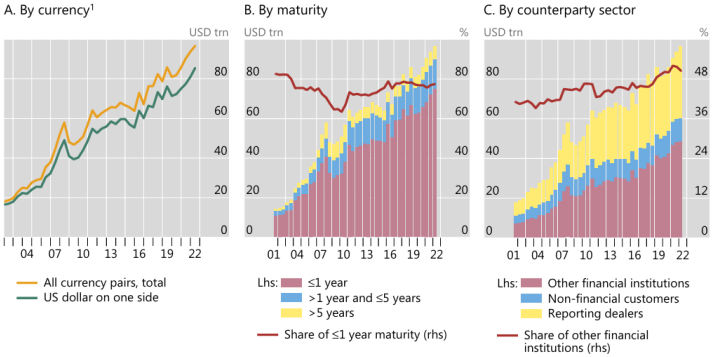
Sources: Hilander (2014); Rush et al (2013); ECB; Eurosystem Working Group on Securities Statistics; The Life Insurance Association of Japan; Barclays; BIS international banking statistics and debt securities statistics; authors' calculations.

Source: BIS.

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Most FX Swaps Involve the USD, Are Short-Term

[Core Concept]



[1] The gold line is the aggregate of FX swaps, FX forwards and currency swaps. The green line is contracts in which US dollars are exchanged.

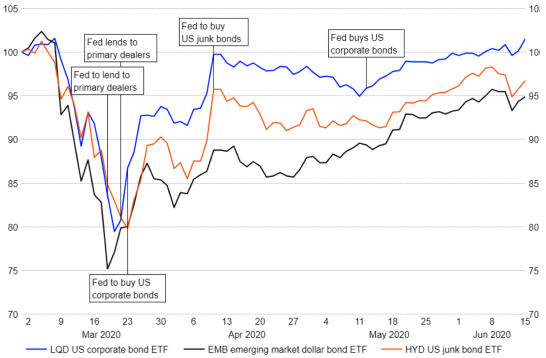
Source: BIS OTC derivatives statistics.

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Fed Intervention in 2020: Bond Prices (McCauley 2024)

[Core Concept]

Figure 5: Emerging Market Bond Prices Rise on Fed US Corporate Bond Buying in 2020



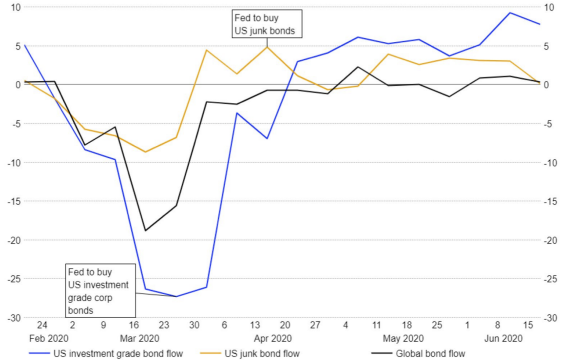
Note: Data depict indices of ETF prices that are indexed so February 28 = 100.
Source: Refinitiv Datastream, Federal Reserve, and author's calculations

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Fed Intervention in 2020: Bond Flows (McCauley 2024)

[Core Concept]

Figure 6: Global Dollar Bond Outflows Stop on Fed US Corporate Bond Buying in 2020



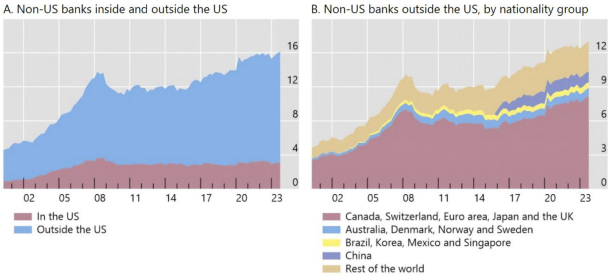
Note: Data depict billions of dollars per week.
Source: Investment Company Institute

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USD Liabilities of Non-US Banks (McCauley 2024)

[Core Concept]

Figure 2: Dollar Liabilities of Banks Headquartered outside the United States

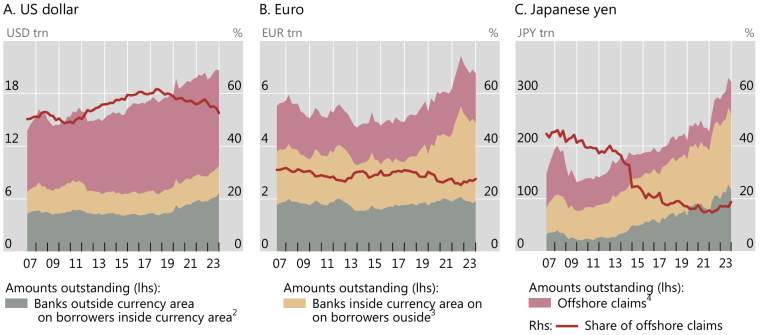


Note: Data depict cross-border and local liabilities in all instruments vis à vis counterparty countries. Data exclude intragroup positions but include liabilities to other (unaffiliated) banks. From the end of 2015, data include positions reported by China and Russia (the latter up to the end of 2021). Data are in trillions of dollars.
Source: BIS consolidated banking statistics, BIS local banking statistics, and author's calculations

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Cross-Border Claims by Currency

[Core Concept]



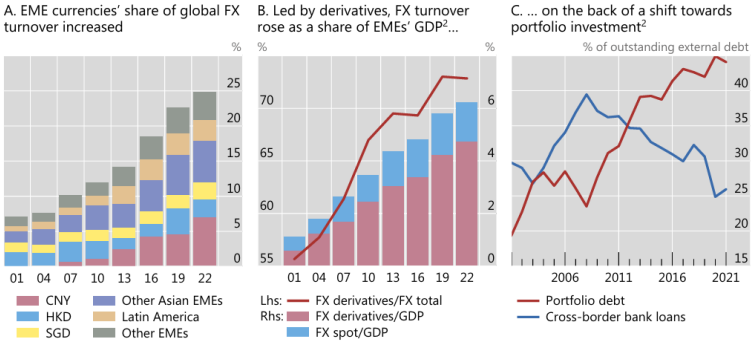
[1] International claims comprise cross-border claims and local claims (on residents) in foreign currencies (including intragroup claims). Panel B excludes intra-euro area cross-border claims. [2] Cross-border claims on residents of the currency area reported by banks abroad. [3] Cross-border claims booked by banks in the currency area on counterparties outside the currency area. [4] Claims booked by banks located outside the currency area on counterparties outside the currency area.

Source: BIS.

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Increase in EM FX and Portfolio Investment

[Core Concept]



[1] See technical annex for details. [2] Median of EME currencies.

Sources: IMF, *Balance of Payments Statistics*; IMF, *World Economic Outlook*; national data; BIS Triennial Central Bank Survey; authors' calculations.

Source: BIS.

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Project NPV, USD M	Unhedged	Hedged
Before the shock	11.306	10.868
Case A: no usable credit	9.872	9.422
Case B: full usable credit	11.021	10.541

- A hedged firm hedges the frozen Year 3 amount at $F_{0,3} = 1.13$. The project stays positive in every case.
- Mid rates: borrow USD 11M, repay $11(1.05) = 11.55$, and buy it forward for $11.55/1.122 + 0.05 =$ EUR 10.3441M. The cost is **3.441%**, below the 3.75% local loan.
- Executable quotes: borrow $10(1.102) =$ USD 11.02M and repay $11.571/1.12 + 0.05 =$ EUR 10.3813M. The cost is **3.8125%**, so the local loan is cheaper.
- Financing enters the valuation once: keep interest out of project cash flows. The funding saving is a separate financing gain.

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